



V(3rd Sm.)-Computer Sc.-G(GE/CC-3)/CBCS

2021

COMPUTER SCIENCE — GENERAL

Paper : GE/CC-3

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions : 2×5
 - (a) State the role of program counter.
 - (b) Write down any two characteristics of IAS Computer.
 - (c) Define the functions of CAM.
 - (d) Write down the significance of cache memory.
 - (e) Convert the following arithmetic expressions from infix to reverse polish notation :
$$A + B * C/D - E$$
 - (f) Write down the two differences between memory mapped I/O and I/O mapped I/O.
2. (a) Define Instruction Cycle. Explain its operation with a flowchart. 3+5
(b) What do you mean by effective address of an instruction? 2
3. (a) What are the three ways a negative integer number can be represented? Explain how $(-12)_{10}$ can be represented in each case. 3+5
(b) Define virtual address. 2
4. (a) Evaluate the following arithmetic expression using three address and one address instruction format :
$$X = P + Q - R - S / M$$
 3+4
(b) Describe the working principle of keyboard. 3
5. (a) Consider a direct mapped cache of size 32KB with block size 512 Bytes. The size of main memory is 16 MB. Find number of bits in tag and line number. 3+3
(b) Write down the differences between CISC and RISC processors. 4
6. (a) Write down the flowchart / algorithm of Addition and Subtraction with sign-magnitude data. 6
(b) Write down the differences between direct and indirect addressing mode. 4

Please Turn Over

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| 7. (a) In how many ways DMA controller transfers the data? Define each of them. | 1+5 |
| (b) Write down the significances of Polling. | 4 |
| 8. (a) Write down the key features of SCSI & PCI bus. | 5 |
| (b) Compare between write through and write back methods in cache. | 5 |
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